Demo reading material, not the paper's original text

Attention Is All You Need

Reading notes written for the PaperDigHelper demo

Why this paper matters

Before 2017 the strongest systems for translating and modelling text read a sentence one word at a time, passing a hidden state along a chain. That made them slow to train, and information from a distant word had a long way to travel. This paper proposes a different design, the Transformer, in which every layer is built from attention and small feed-forward networks.

The core idea

Attention lets a word put a question to the other words in the sentence and take a weighted mix of their answers. Self-attention lets every position look at every other position in a single step, so two distant words are connected by a path of length one. Because all positions are handled together, the whole sentence can be processed in parallel on modern hardware, which is a large part of why training became so much faster.

Order and several views

Attention by itself ignores word order, so the model adds a position signal to every word before the first layer. The paper uses fixed sine and cosine patterns of different wavelengths for this signal, and reports that learned position vectors worked about as well.

Multi-head attention runs several small attention computations side by side, so different heads can follow different relations in the same sentence. One head may track which noun a pronoun points to while another follows the words nearby.

What the results showed

On English-to-German and English-to-French translation the Transformer reached higher scores than the best earlier systems at a fraction of their training cost. The same architecture also worked for English constituency parsing, which suggested it was not tied to translation alone.

Limits to keep in mind

Attention compares every position with every other one, so its cost grows with the square of the sequence length. Very long documents therefore need cheaper variants, which is why many later papers in this library are about efficient attention.